Feature Engineering Analysis

The first feature I incorporated was Length, creating the subfeatures for the length of the run in words, length of the run in characters, and length of the question in terms of the length of the tokenization list. Other options that were considered but not ultimately included were the length of the guess in chars and the length of the guess in words.

After addition of each feature, I noted the new Buzz ratio and "best" percentage of that feature, alone and/or in combination with the existing "best" features. I found that adding the length of the guess in chars, with the length of the run in words, length of the run in chars, and the length of the question in terms of length of tokenization list, reduced the local data's Buzz ratio by 3.00 and best ratio by .01. I also found those features plus the length of the guess in words decreased the Buzz ratio by 2.50 and the best ratio by .01. As such, for the Length feature, the best subfeatures were determined to be the length of the guess in chars, with the length of the run in words, length of the run in chars, and the length of the question in terms of the length of the tokenization list.

Next, I compared the addition of the Category and Subcategory features. Both Category and Subcategory alone produced a Buzz ratio of 57.00 and best ratio of .4. However, Category and Length produced a higher Best ratio than Subcategory and Length. This combination of features resulted in "good enough" improvement over the baseline results on the test data in gradescope, which aligned with the generalizations I made on the training data given to us to use locally. The final weights of each category were between -1 and 1, showing that they were less impactful on the final classification than Gpr confidence but still significant. Moreover, the category weights were generally greater (in absolute value) than the Length subfeatures, showing that they impacted the final classification more than Length. Finally, the signs of the category weights (positive/negative) reflect trends observed in the training data, as semantically, these categories would not indicate to humans how to determine the correctness of an answer.